

DESIGN STUDIES OF FAST REACTORS IN FRANCE

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1. Introduction

Several years ago while Rapsodie, the first fast reactor in France, was being manufactured and built, studies were begun on fast reactors of the future.⁽¹⁾ At the beginning of 1965 efforts were concentrated on the design study of the prototype reactor Phenix, construction of which is expected to start in 1969. It is now decided that Phenix will be a sodium cooled reactor using mixed uranium and plutonium oxide as fuel. Although the power has not been defined yet, studies will be based on a value of 250 MW(e). Current thinking and work on the overall design of the reactor and the primary cooling circuits will be outlined here. The reliability and safety of the installation depends to a large extent on the design of the whole and on the type of equipment used. This is why attempts have been made to assess the respective merits of two fundamental alternatives:

- (a) a reactor contained in a vessel with external primary circuits, which will be referred to as the conventional design, and is used, in particular, for Rapsodie;
- (b) a reactor with integral primary circuits, i.e. immersed in a large vessel filled with sodium. The American reactor EBR-II is the first to have paved the way in this direction.

These two possible designs were considered and studied simultaneously while efforts were concentrated on the integral circuit system which was the more novel. Prior to, or during, this work attempts were made to define the values of the operating temperatures—steam conditions, primary and secondary sodium temperatures.

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2. Choice of Steam and Sodium Temperatures

It became apparent that a general optimization attempt, taking into account not only the operating parameters but also the structural parameters, and in particular the geometrical and physical characteristics of the core, would be too ambitious. Although this study has been started, for the moment a partial optimization has been sufficient, relating to the thermal operating characteristics of the installation.⁽²⁾

In this study,* various possible cores were considered, each being characterized only by its pressure drop and by a law defining the fuel can temperatures, taking into account the hot spot factors. Several possible pressure drop values were therefore considered (4, 5, 8 and 10 bars for 200°C heating of the sodium), as well as an approximate and provisional law linking the hot spot temperature of the cans with the sodium temperatures:

$$T_g = T_s + 0.8 (T_s - T_e) + 8,$$

T_g , T_s and T_e being respectively in °C the can hot spot temperature, the temperature of the sodium leaving the reactor, and the temperature of the sodium entering the reactor.

This law would approximately correspond, for Phenix, to the pessimistic case of almost complete absence of sodium mixing around the fuel pins. Calculations have been made for different values of T_g , in particular: 650, 675, 700, and 750°C.

Preliminary studies first showed that:

- (a) Completely feasible sodium characteristics (about 600°C at exit from the reactor, temperature rise less than 200°C) would correspond advantageously to a supercritical steam cycle, which will have to be considered for the fast reactors of the future.
- (b) As this type of cycle appears premature at present, it was necessary to use a steam cycle similar to that of the present conventional French stations.† The superheated steam pressure was therefore fixed at the highest value compatible with present French technology: 163 bars (166 kg/cm² absolute).
- (c) A reheat cycle gives a 0.94% increase in efficiency, which greatly offsets the additional investment it requires. This type of cycle is essential anyway if one wishes to use a conventional turbo-alternator set.

Bearing in mind the choice which has been mentioned (steam pressure, feedwater temperature, can hot spot temperature law), a calculation was

* Stein & Roubaix contract.

† Steam (at turbine): superheat 565°C 163 bars absolute
 reheat 565°C 33.9 bars absolute

Feed-water: 252°C.

made, for different superheat temperature values (515° , 540° , 565°C), of the primary and secondary sodium temperatures which give the minimum total expenditure corresponding to the variable charges:

- (a) investment costs;
- (b) fluid circulation costs;
- (c) energy loss (greater or lesser efficiency of the cycle).

It has been found that the minimum total cost is obtained for a superheat temperature of 540°C .

The optimum operating conditions, for a can hot spot temperature of 700°C and a pressure drop in the core of 6.5 bars approximately, are given below:

Steam:	superheat and reheat	540°C
	superheat pressure	163 bars
	feedwater	289°C
Secondary sodium:	$556\text{--}348^{\circ}\text{C}$ (temperature rise 208°C)	
Primary sodium:	$567\text{--}411^{\circ}\text{C}$ (temperature rise 156°C)	

These results will have to be adjusted in accordance with the exact design of the reactor and above all by using more definitive data on the characteristics of the core. It is unlikely that the actual thermal characteristics of Phenix will differ much from the values which have just been given.

3. Design of the Reactor and the Primary Circuits

As has already been mentioned, two possible systems were studied simultaneously and compared—reactor with external circuits and reactor with integral circuits. The integral primary circuit system necessarily means that the neutron shields are themselves immersed in the sodium, since they must be situated between the reactor and the intermediate exchangers. The external circuit system, on the other hand, can accept a neutron shield inside the reactor vessel as well as a shield brought to the outside of the vessel.

In fact a number of systems had to be considered, from the integral circuit design to that with external circuits and external shielding, including intermediate alternatives such as: semi-integral system (exchangers in the reactor vessel, pumps outside), or external circuit system with neutron shielding inside the vessel.

Only the salient points of the studies carried out will be given, with emphasis on the integral circuit concept which is less conventional.

Reactor with External Primary Circuits⁽³⁾

The number of circuits envisaged for Phenix (600 MW(th)) is three, taking the following considerations into account. Two circuits constitute the indispensable minimum to ensure cooling in the event of accidental shutdown

of one circuit. However, to be able to operate at reduced power in the event of failure of one circuit, the above remark means in fact that the number of circuits to be provided is at least three. As a greater number would lead to a higher construction cost and to components which were less representative of those of a future station, the choice of three circuits seems the optimum, each circuit being fitted with a pump and an exchanger. In the case of a reactor of 1000 MW(e) (about 2400 MW(th)) the number of primary circuits would probably be four.

Two alternatives were considered, in relation to the sequence of passage through the equipment: the sodium can pass successively through the reactor, the pumps, and the exchangers (pumps on the hot spot), or on the other hand through the reactor, the exchangers, and the pumps (pumps on the cold spot). As the pumps envisaged are centrifugal vertical pumps with a free level, the difference in the sodium levels in the reactor and in the pumps represents the pressure drop between the exit from the vessel and the pumps. The system with pumps on the hot side obviously reduces this pressure drop through the absence of the exchanger on the reactor-pump section: the shaft of the pumps is shortened by this. However, as the intermediate exchanger is then on the delivery side of the pumps, it would undoubtedly mean giving the secondary sodium a pressure adequate to avoid the active primary sodium passing into the secondary circuits, in the event of a leak in an intermediate exchanger. It finally turned out to be preferable to place the primary pumps on the cold side, i.e. downstream of the exchangers.

It is then necessary to define the broad outline of the intermediate exchangers. These units must be capable of repair in the event of a leak. We thought of either units with tube bundles which can be dismantled, or U-shaped units which cannot be dismantled, which simplifies the design and construction. In the event of a leak in the unit it is possible to gain access to it after shutdown of the reactor and isolation of the circuit to block the damaged tube by plugging it.

After selecting the number of primary circuits, establishing the type of pump and their position in the circuit, and defining the type of intermediate exchanger, the important thing then is to obtain the simplest possible layout of the circuits. Two methods are possible depending on the design of the neutron shielding; shielding housed in the reactor vessel or shielding located outside the vessel.

If the neutron shielding is internal, its design must be more elaborate (borated graphite in stainless steel boxes) and its price higher. It has the advantage, however, of confining the neutrons in the most restricted volume possible and thus permitting a simple layout for the circuits, without the trouble of neutron leakage problems through the sodium ducts. It is useful to use ducts which are as short and straight as possible. It is, however, necessary to absorb the expansion of the pipes by means of suitable expansion

bellows or by designing the pump and the exchanger of each circuit in the form of an elastically mounted unit to allow it horizontal movement.

Figure 1 gives an example of a design for a 1000 MW(e) installation corresponding to this arrangement. It will be seen on this drawing that fuel handling is provided by a mechanism introduced laterally at the top of the vessel (rotary crane in the reactor vessel).

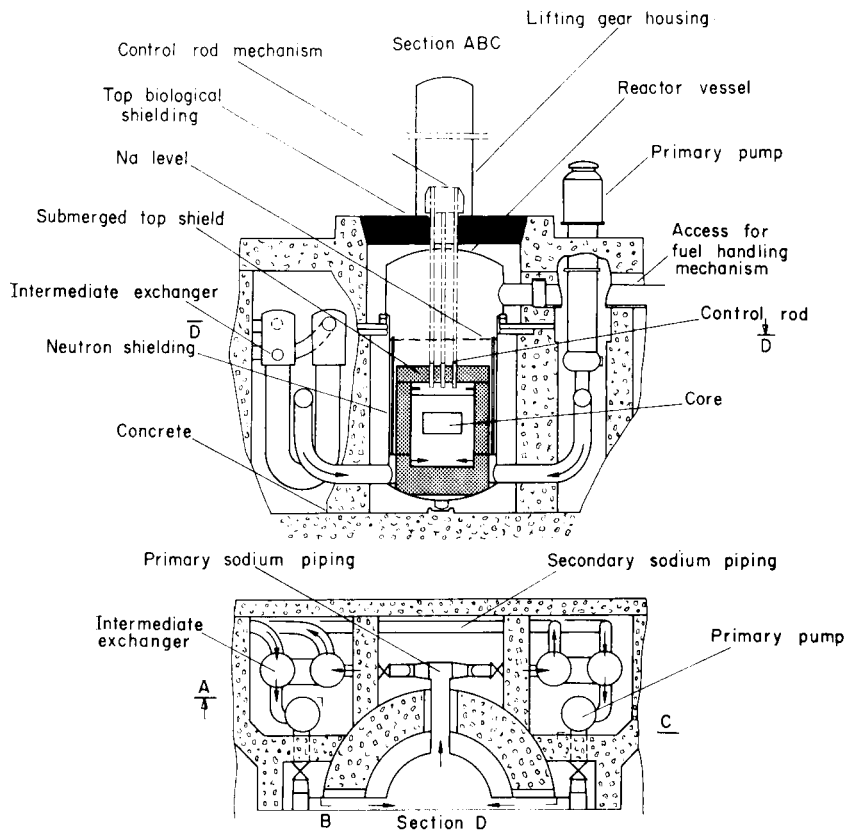


FIG. 1. Reactor with external circuits and integral neutron shielding.

If the neutron shielding is essentially outside the reactor vessel, the diameter of the vessel is reduced to its minimum value; it contains only the core and blanket assemblies, the necessary internal storage for cooling of the spent assemblies before discharge, and steel hoops limiting the neutron flux on the vessel. On the other hand, the neutron shielding outside the reactor must be designed with care. The inevitable clearances in the upper biological shield slab may require additional shielding blocks forming covering plates

at ground level; the sodium ducts also form outlet holes for the neutrons, which must be taken into account for the shielding of the intermediate exchangers. Amongst the different possible systems, it seems that the most logical one which also satisfies the free expansion requirements of the piping

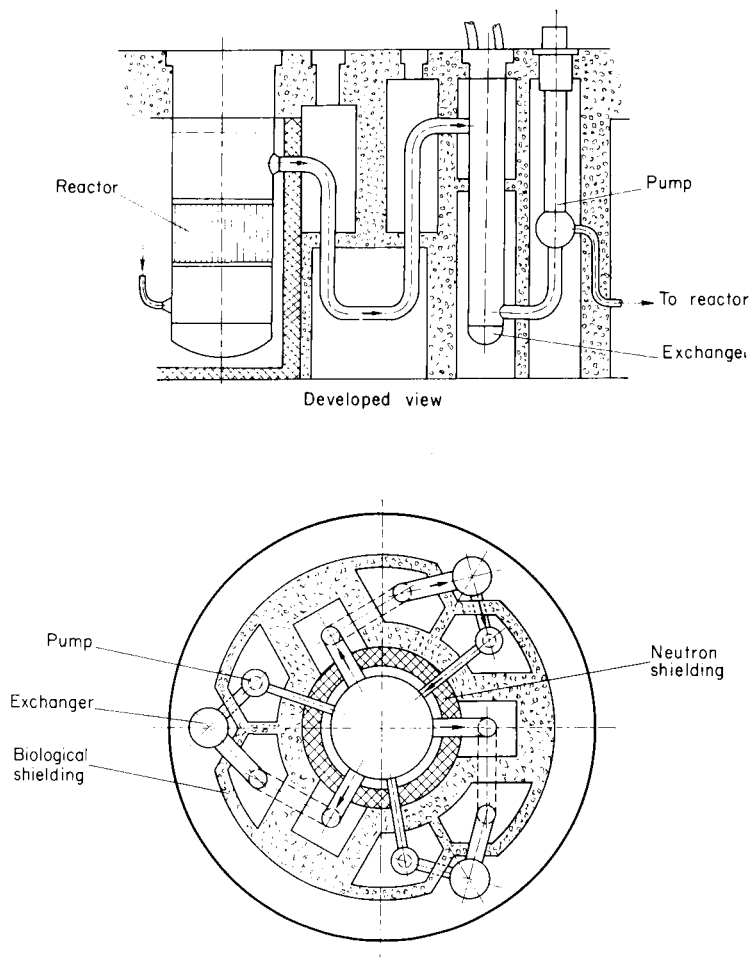


FIG. 2. Reactor with external circuits and non-integral shielding.

is to use a fairly winding route between the vessel and the exchangers, passing through the successive chambers with concreted walls. The layout of the return piping between the pumps and the exchangers may be simpler, if very slight activation of certain elements in the pumps is accepted. Figure 2 gives a simplified diagram of the design of a Phenix reactor based on the above considerations.

Study of the design of a high power fast reactor, with the system of external primary circuits, revealed the following characteristics:

- (a) the diameter of the vessel is limited: of the order of 5 m for a reactor of 250 MW(e), of the order of 6.50 m for a reactor of 1000 MW(e);
- (b) the layout of the piping can be simple in the case of shielding inside the vessel, if expansion bellows are used on the pipes, or if provision is made for possible horizontal movement of the pumps and exchangers. The layout is more complex if the neutron shielding is outside;
- (c) external circuits may facilitate any access to the equipment (pumps and exchangers in particular) for repair or maintenance operations;
- (d) reliability of the cooling system in the event of a leak means providing various devices such as: double wall around part or all of the primary installation (depending on whether there are isolating valves or not), siphon-breaker with an auxiliary cooling circuit independent of the main circuits.

Reactors with Integral Primary Circuits

The only reactor of this type built at present is EBR-II. The idea has been taken up, with a very different design, in the study of a reactor of 1000 MW(e) undertaken by General Electric. In both cases the reactor is contained in a closed vessel housed in the sodium vessel, the cover of which has to be removed for fuel handling operations.

Another alternative of the system has been studied with integral primary circuits, in which the reactor is freely open in the sodium vessel.* The fuel charging and discharging operations are therefore not fundamentally different from those in a reactor with external circuits and do not involve any prior opening operation on an internal vessel containing the reactor.

The design envisaged (Fig. 3) is described below,^(4, 5) the dimensions given corresponding approximately to a reactor of 600 MW(th) (Phenix).

The reactor vessel, called the primary vessel, contains the core, the radial and axial blankets, the radial neutron shielding, the bottom neutron shielding and possibly a top neutron shielding. At the top the vessel widens to act as a substructure for the pumps and exchangers. The vessel is therefore made up of two cylindrical hoops of different diameter, connected by a horizontal collar. The bottom hoop which contains the reactor proper is about 5 m in diameter and 8 m high; it is closed at the bottom by a flat or dished end. The top hoop is about 9 m in diameter and 1.5 m high; it is open at the top. The horizontal collar which joins the two cylindrical parts is penetrated by circular holes for the exchangers and the pumps. The holes for the exchangers simply provide a passage for the tube bundles of the units. The pumps are inserted

* Recent information has revealed that the future British PFR reactor is of a design similar to that described below.

in individual cylindrical hoops open at both ends and welded at the bottom to the collar of the vessel.

The primary vessel, the pumps and the exchangers are contained in another vessel about 10 m in diameter and 12 m high, filled with sodium, called the

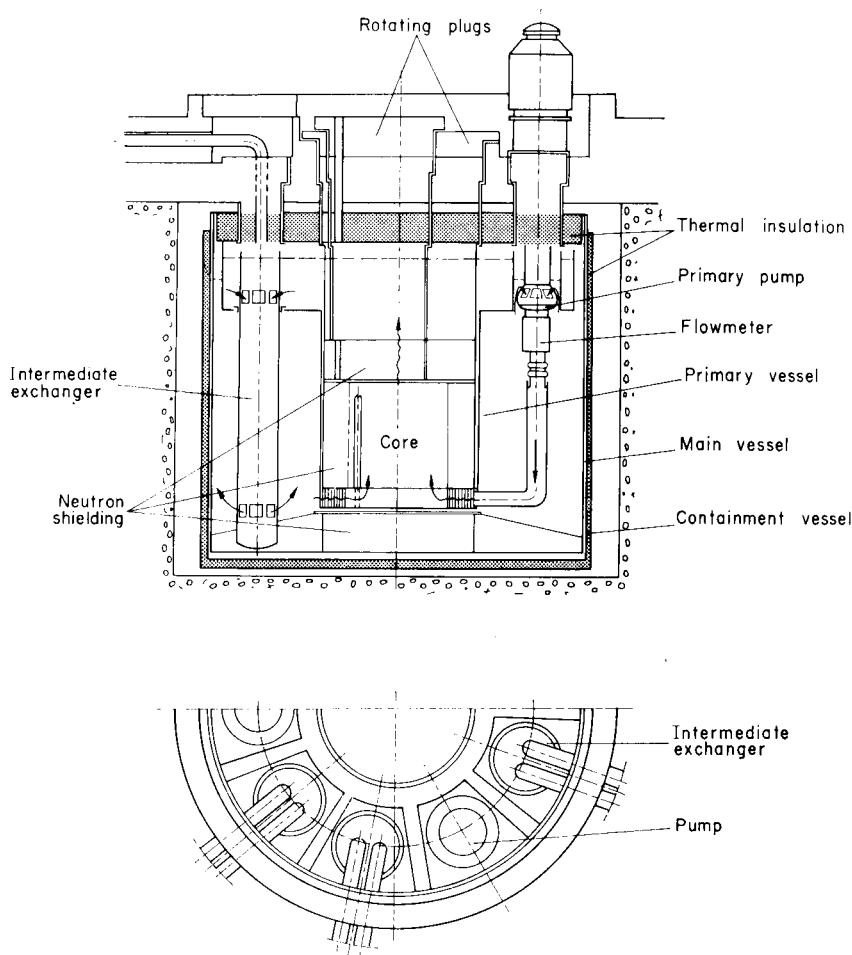


FIG. 3. Phenix (integral circuit version).

main vessel in the following. The top face is flat and fitted with extensions through the biological shielding to house the shield plugs for the reactor, the pumps and the exchangers. A containment vessel is placed around the main vessel to form a double shell.

The exchangers are similar to those of Rapsodie, except of course that there is no longer an outer shell to contain them, apart from at the top (extension

of the vessel into the shielding). They are made up of a bundle of vertical tubes which can be dismantled, encased in a sliding sleeve which can also be dismantled with the bundle.⁽⁶⁾

The pumps⁽⁶⁾ are similar to those of Rapsodie, i.e. centrifugal, axial delivery and capable of being dismantled. The fixed outer shell comprises here, on the one hand, the extension of the main vessel into the shielding, and on the other, the hoop open at both ends, welded on the horizontal collar of the primary vessel. The delivery side of the pump is connected by a sliding connexion to a pipe leading to the supply header for the assemblies. It will be seen that this is the only element of piping in the primary installation.

The sodium circulation through the system can be deduced from the above description: the sodium passes up the core and blanket assemblies, through the top neutron shielding and flows into the top of the primary vessel, then drops in the intermediate exchangers and arrives in the main vessel; the pumps draw the sodium from this vessel and deliver it into the pipes joining the bed (fuel assembly supply header).

A single argon atmosphere is provided above the sodium, the primary vessel opening freely at the top into the main vessel. It follows that the free surfaces of the sodium are at two levels during operation; one level in the primary vessel and a lower level in the main vessel and in the hoops for the pumps. The first level characterizes the pressure at the outlet from the reactor, the second at the outlet from the exchangers (pump intake). The difference between the levels is of the order of 0.6 m.

We note finally that the main vessel only contains the reactor and the primary "circuits" proper. The ancillary circuits are left outside, in particular those for filling, purification, burst can detection. The corresponding small diameter pipes enter the main vessel near the sodium level.

In the main vessel there must be the equipment necessary for neutron measurements, flow measurements, temperature measurements, and level measurements.

The neutron measurement chambers are contained in vertical sheaths lagged internally and provided with internal gas cooling (this is the system used in EBR-II).

The flow meters are placed immediately on the delivery side of the pumps and can be dismantled with them.

The essential characteristics of the proposed system are as follows:

- (a) The extreme compactness of the installation results in an appreciable saving in the general overall dimensions.
- (b) Circulation of the sodium is ensured essentially by the arrangement and the design of the primary and main vessels. The primary piping is limited to a length of a few metres for each pump, instead of about 50 m per circuit in the case of a conventional design. On the other hand,

the diameter of the main vessel is large: 10 m at least, instead of about 5 m for a reactor with external circuits. It follows that the vessels must be constructed on site and designed with care.

- (c) Preheating of the installation is simple as a result of its compactness. It can be achieved by circulation of hot argon inside the main vessel.
- (d) The internal structures of the main vessel (primary vessel and pump delivery pipes) do not require perfect leak-tightness.
- (e) The internal neutron shielding (which raises its own problems anyway) lightens the top shield slab, which is nothing more than a gamma shield. This slab becomes thin and homogeneous and the problems of neutron leakage through the plug clearances no longer arise.

In addition, from the point of view of safety and reliability:

- (f) The simplifications in the design and construction must limit operating incidents and the resultant maintenance work.
- (g) The failure of a pump or a primary exchanger results solely in a reduction of the flow or of the exchange surface, but does not necessitate shutdown of a complete cooling circuit.
- (h) The great mass of sodium provides substantial thermal inertia in any cooling incident. It attenuates the thermal shocks, from wherever they come.
- (i) Reliability of the cooling system in operation or after shutdown (with the pumps assisting, if necessary) depends only on a permanent level in the main vessel. In view of the large volume of sodium, a drop in level following a leak in the main vessel could only be very slow. For the same reason the containment vessel can easily guarantee a very small ultimate drop in level without any effect on the sodium circulation.
- (j) Any ancillary piping (purification circuit) can enter the vessel near or above the sodium level. The part of the vessel immersed in the sodium thus remains a perfect cylinder, with nothing which can cause any perturbation or leak.
- (k) The immersed part of the vessel is at the temperature of the cold sodium and without any pressure.

To the basic description given above will now be added certain information regarding the structural problems.

Materials

The whole of the installation, in particular the primary and main vessels, will be of austenitic stainless steel, as a rule grade 316. The question of whether the containment vessel can be profitably built of ferritic steel will be examined; this does not appear to be at all certain.

Supports for apparatus and vessels

The problems of supporting the vessels and the main equipment (pumps and exchangers) are to a great extent conditioned by the questions of expansion and relative movements which may ensue.⁽⁷⁾

For the main vessel, and the containment vessel which surrounds it, two systems of support have been studied, but the choice between them has not yet been made:

- (a) Vessels resting with their bottoms on the external concrete structures; the vertical expansions would then be absorbed by expansion bellows at the top, under the biological shield slab. The primary vessel would also be fitted in the main vessel.
- (b) Vessel suspended at the top on a metal framework incorporated in the shielding concrete; the vessels would then be free to expand downwards. The primary vessel could either be fitted in the main vessel or suspended by its end on the upper framework.

In all cases the roof of the vessel, i.e. the horizontal sheet which closes off the vessel under the top shielding slab, is kept cold (about 60°C) by internal lagging and conditioning of the outside atmosphere. The pumps and exchangers are fixed into the upper slab.

The relative movement (due to expansion) of the collar of the primary vessel with respect to the exchangers is possible because of the floating sleeve surrounding the tube bundle, the support collar of which permits suitable horizontal movement (of the order of 3 cm).

The (fixed) pumps are connected to the delivery piping (which can move as a result of expansion) by means of a sliding joint. A bellows on the delivery side allows the horizontal movement to be absorbed.

Number and arrangement of pumps and exchangers

A preliminary study was carried out to define the optimum number of units, taking the following criteria into account:⁽⁸⁾

- (a) reliability and load factor of the station;
- (b) construction costs;
- (c) size of the equipment.

It was found that the most favourable selection was:

- (i) three pumps and six exchangers for a 250 MW(e) reactor;
- (ii) three or four pumps and six or eight exchangers for 1000 MW(e).

The exchangers and the pumps are arranged symmetrically around the vessel. For example, in the case of Phenix, the exchangers are arranged in three sets of two, separated from one another by a pump. In this way, in all

cases of operation, in steady state and transient conditions, we obtain perfect symmetry in the sodium flow and thus in the temperatures of the vessels or the equipment.

Neutron shielding

The lateral and top shielding is essentially made up of blocks of borated graphite contained in stainless steel leak-tight jackets. The side shielding at the bottom of the fuel assemblies might possibly be of boron carbide. The bottom shielding is of borated steel.

The geometrical structure envisaged is as follows: the bottom shielding would be a block of horizontal plates of borated steel, separated by layers of sodium, and situated below the supply header for the assemblies. The side shielding would be made up of vertical elements fairly similar to the fuel or fertile assemblies but of circular cross-section (a hexagonal shape being more complicated, weaker, and unnecessary); the bottoms of the shield assemblies would be either of borated steel or of boron carbide. The top shielding can be in the shape of a metal tube block, the tubes being filled with borated graphite and the sodium leaving the core circulating vertically around the tubes. If handling is carried out by means of rotating plugs, this block would be made up of two cylindrical eccentric parts with the same dimensions as the two top plugs to which they would be rigidly attached, and of which they would to some extent form an integral part. If handling is effected by means of arms, the shield block could be raised to permit access by the arms to the tops of the assemblies.

A top shield made up of elements incorporated in each fuel or fertile assembly is also being considered. This system has its advantages and drawbacks. It has at all events one great merit; for internal transfer of the assemblies it does not affect handling at all, except through the elongation of the assemblies.

4. Control Rod Mechanisms

Operation of the control rods from the bottom of the reactor was rejected in the case of Phenix, since no major advantage could be found over operation from the top, which is a simpler and more conventional system.

The design of the rod mechanisms can be greatly affected by the choice of fuel handling system. For example, straight mechanisms after the pattern of those used on Rapsodie can be envisaged if fuel transfer is achieved by two rotating plugs or by an arm in a rotating plug. If the transfer were achieved by one arm alone, the rod mechanisms would either have to be retractable or offset with respect to the control rods.

For reasons of reliability and safety the system with straight mechanisms is definitely preferred controlling the rods from directly above them. The

characteristics of the mechanisms would not then be very different from those of Rapsodie.

5. Fuel Handling

In a high power fast reactor it is necessary to remove the after-heat of the fuel assemblies for several months before being able to transfer them to final storage. Consequently the progress of a spent fuel assembly which is ready to be taken out of the core must take place in the following stages:

- (a) Transfer of the assembly from the core to temporary storage in which adequate cooling takes place. It is convenient for this storage to be located inside the reactor vessel where the sodium circulation permits easy removal of the after-heat.
- (b) Discharging of the assembly after several months' cooling in the internal store: the assembly is taken out of the reactor vessel and transported to an external store.
- (c) External handling: the assembly is cleaned, then sent to the cutting-up and processing installations.

The handling mechanisms, the design of which is linked with that of the reactor itself, are those for the first two stages mentioned above: internal transfer and discharge. It is these operations and the corresponding mechanisms which will be discussed below.

For Phenix, annular internal storage was chosen: the assemblies withdrawn from the core are stored in a circle around the radial blanket. This is the simplest arrangement and the easiest to operate. The rule was also laid down that the assembly would always be immersed in the sodium in the vessel during its transfer. This is the only simple and safe way to guarantee permanent cooling.

Before discharge, the assembly will be brought into one or two special positions in the vessel called discharge stations. As a result there will be one or two discharge channels in the top shield occupying fixed positions for the discharge of all the assemblies. It is intended to place a discharge pot at these discharge stations: the assembly will be placed in the pot and withdrawn from the vessel in the sodium filled pot. The power of some 3 kW which is still being emitted will thus be more easily absorbed or removed.

As regards the transfer mechanisms, different possible systems were examined to compare the respective merits, in particular:

- (a) a set of two rotating plugs (system used for Rapsodie): large plug centred on the axis of the reactor, small plug offset, inside the first one;
- (b) an arm, either articulated or not, housed in a rotating plug centred on the axis of the reactor or offset;

- (c) an articulated arm in the axis of the reactor, or in three positions at 120° from one another.

In the case of the rotating plugs there is benefit of 2 years' favourable experience of testing and operation on the Rapsodie mockup vessel. In the case of the system with two rotating plugs, however, the large plug would be more than 5 m in diameter for Phenix (that in Rapsodie is less than 3 m), for a reactor of 1000 MW(e) the diameter would exceed 6 m, which would probably lead to on-site fabrication.

The use of folding arms without rotating plugs would mean that, for movement of the arm, the space above the assemblies must be entirely free and consequently the control rods would have to be concealed in the top shield. This withdrawal of the rod mechanisms appeared as an important inherent complication in this system.

In the case of the system with an arm and a rotating plug two alternatives were examined, either with the rotating plug centred on the axis of the reactor (the only movement needed from the arm is then a radial extension), or with an offset plug (as the only movement required from the arm is rotation, the arm can then be rigid). The tendency is to prefer the second alternative, while providing for possible bending of the arm perhaps, in the event of possible dismantling. The diameter of the rotating plug remains similar to that of the large plug in Rapsodie.

For discharge of the assembly (removal from the vessel) it has already been indicated that it would be in a pot, and in one or two fixed positions (discharge stations). The discharge stations can be located *a priori* on the ring forming the internal store or on a circle of larger diameter. This second possibility is undoubtedly more favourable since it permits operation in a freer area. It will also be seen that it suits the discharge machine proposed below.

The discharge machine envisaged is in fact either a mobile flask or a fixed lock. The mobile flask would be essentially a leak-tight cylinder surrounded by shielding and of suitable dimensions for accommodating the assembly and the gripping and lifting mechanisms. This flask would be coupled to the discharge channel for each operation, then transported to an external store to set down the assembly there. The lock system, on the other hand, would be a completely fixed arrangement consisting of a large leak-tight box of parallelepiped form, in which the mechanisms would provide the discharge pot with the most direct passage possible from the discharge channel to the storage well. The discharge pot envisaged is permanently fixed on a cable and guided by rigid ramps in its movement between the reactor vessel and the discharge lock. This movement would be vertical or oblique, as required. The lock arrangement appears advantageous because it is completely fixed and it therefore avoids, compared with the flask, first, any movement of a

heavy machine, and secondly, any successive coupling to the handling channel and then to the storage well. It should be noted that this system of discharge would not suit the transfer system of two rotating plugs very well. On the other hand it is very suitable for transfer by means of an arm in a rotating plug.

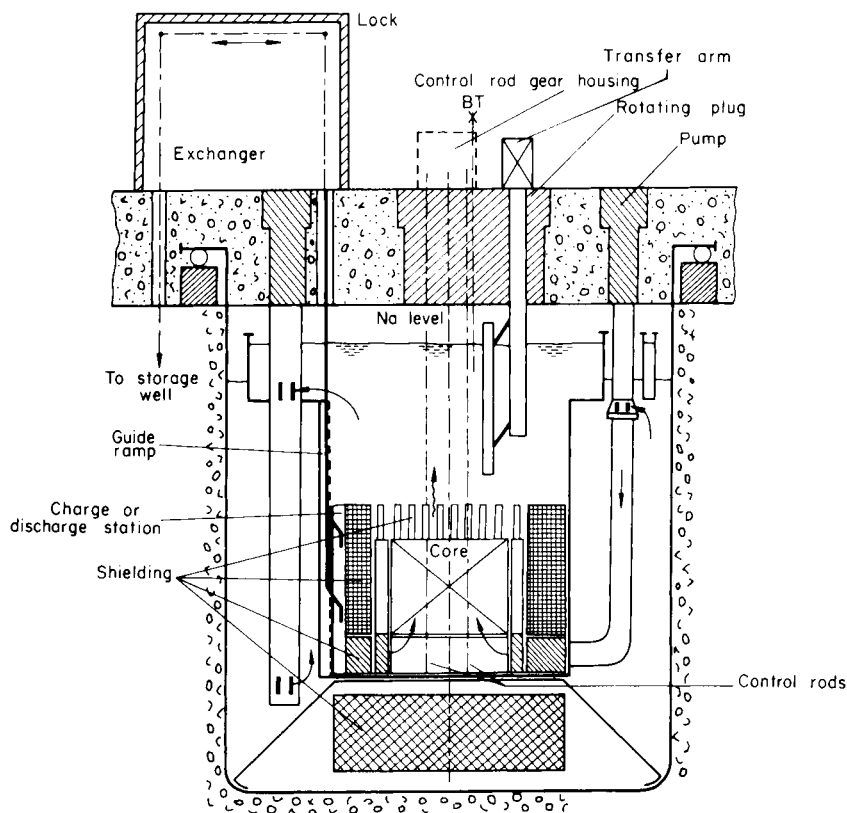


FIG. 4. Transfer by rotating plug (offset) and arm charging and discharging by fixed guide.

As an example Fig. 4 shows the whole of the transfer and discharge systems in the case of an integral circuit Phenix reactor. The combination of a rotating arm and an offset plug (for transfer) with a discharge lock has been shown in this diagram.

6. Conclusion

These are in outline the authors' present ideas on the design of future fast reactors. The final decisions have not yet been taken for the Phenix prototype

reactor. The foregoing pages show the different directions in which studies are being pursued to greater or lesser extents. These studies have been carried out under the Euratom-CEA contract of association.

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